

ABYSSALEARTH

ASSET REVIEW · PHASE 1

Cross-Biome Hero Asset

Geometry Validation Against Concept Art

Conducted June 13, 2026

Reviewers

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Ground truth: the concept art. This document is the GO / NO-GO gate before Phase 2 (full Luminous Rift sweep) and, ultimately, UE5 implementation.

Executive Summary

Phase 1 selects one hero asset from each of the six biomes and rebuilds its Blender greybox geometry so the silhouette reads unambiguously as the concept art intends. These six are the most visually distinctive shapes in their biomes; getting their mass, scale and shape language right de-risks the long tail of supporting assets that follow.

Every prior build shared one of two failure modes. The first is broken geometry: loose vertices authored with no faces (the Glassroot buttress roots, the vein ridges) or chains of unmerged primitives masquerading as a single form (the Fossil Sky spine). The second is wrong shape language at the right-ish scale: a circular pipe where a rectangular gate belongs, fat prisms where sharp crystal spires belong, a flat panel where a leaning stone tower belongs.

Each rewrite is documented against its reference image IDs (recorded in the script docstring under a Concept: tag) and constrained by named UPPER_CASE scale constants so the dimensions are auditable rather than buried as magic numbers. All sixteen CI validation checks pass on the branch, including the new concept-reference, floating-vertex, scale-constant and render-guard checks.

The pages that follow pair each concept image (top) with the Blender wireframe (bottom). The reviewer checkpoints are scale, mass distribution, and distinctive shape language — not surface finish, which is a UE5 material and lighting concern handled at handoff. A GO advances the asset; a NO-GO triggers a geometry refinement round before Phase 2.

Assets under review

- | | |
|--|---------------------------------|
| • Orb Hub — Luminous Rift | Concept: AD-001, LR-006, LR-007 |
| • Glassroot Root Column — Glassroot Forest | Concept: GR-001, GR-002 |
| • Submerged Ruin — Inner Sea | Concept: IS-001, IS-003 |
| • Ceiling Skeleton — Fossil Sky | Concept: FS-002, FS-003 |
| • Portal Ring — Gravity Well | Concept: GW-001 |
| • Mineral Flower — Mantle Garden | Concept: MG-001, MG-002 |

1. Orb Hub — Luminous Rift

Luminous Rift · Concept: AD-001, LR-006, LR-007



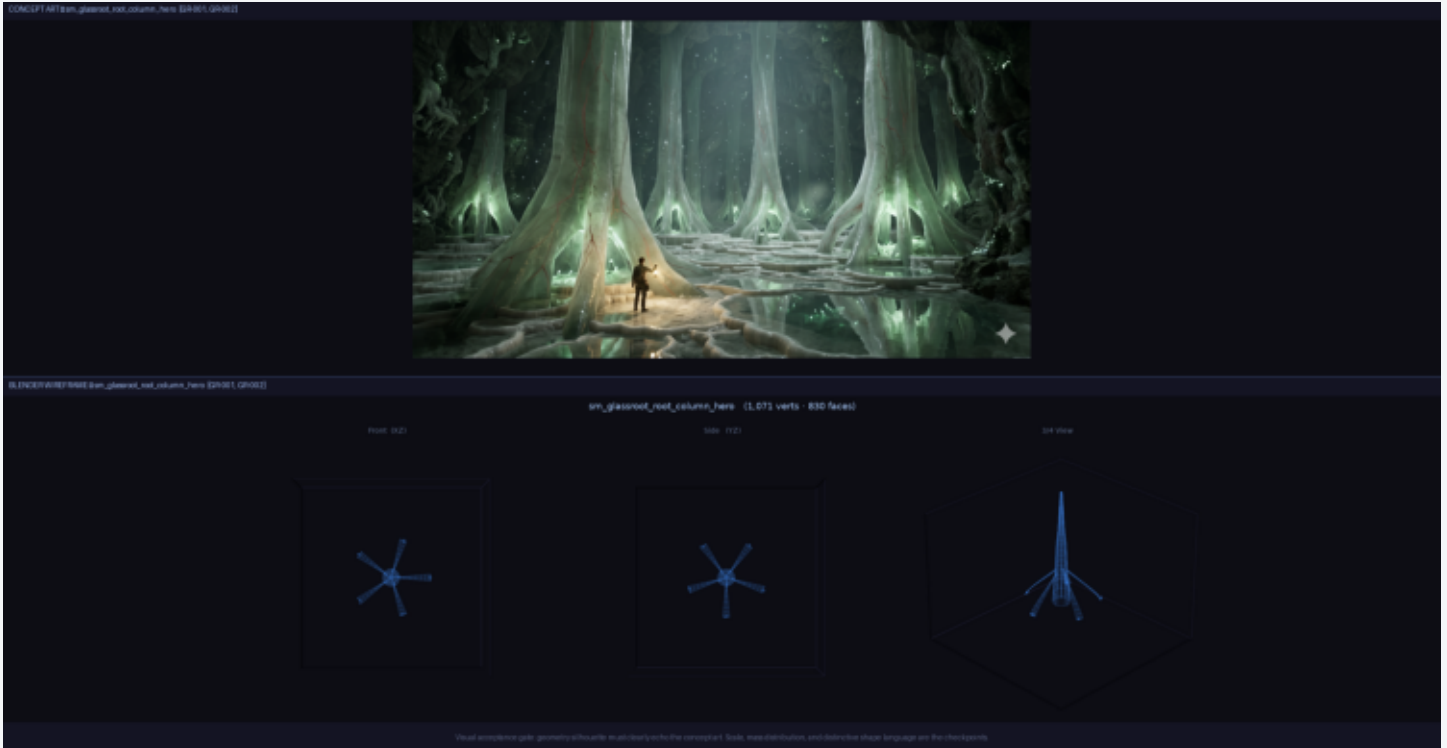
Assessment

The original build produced a 2.8 m sphere with eight wire-thin arms in the wrong plane — roughly a fifth of the intended scale and none of the mechanical mass. The rewrite lays a 76 m annular ring (48-segment tube) in the XY plane, drives eight rectangular box-beam spoke arms inward to a 24 m central orb, and hangs an observation platform below on cable struts. Cable grooves, junction nodes and beam emitter ports restore the heavy machined silhouette AD-001 looks up into.

Verdict: ☐ **GO** ☐ **NO-GO (refine)**

2. Glassroot Root Column — Glassroot Forest

Glassroot Forest · Concept: GR-001, GR-002



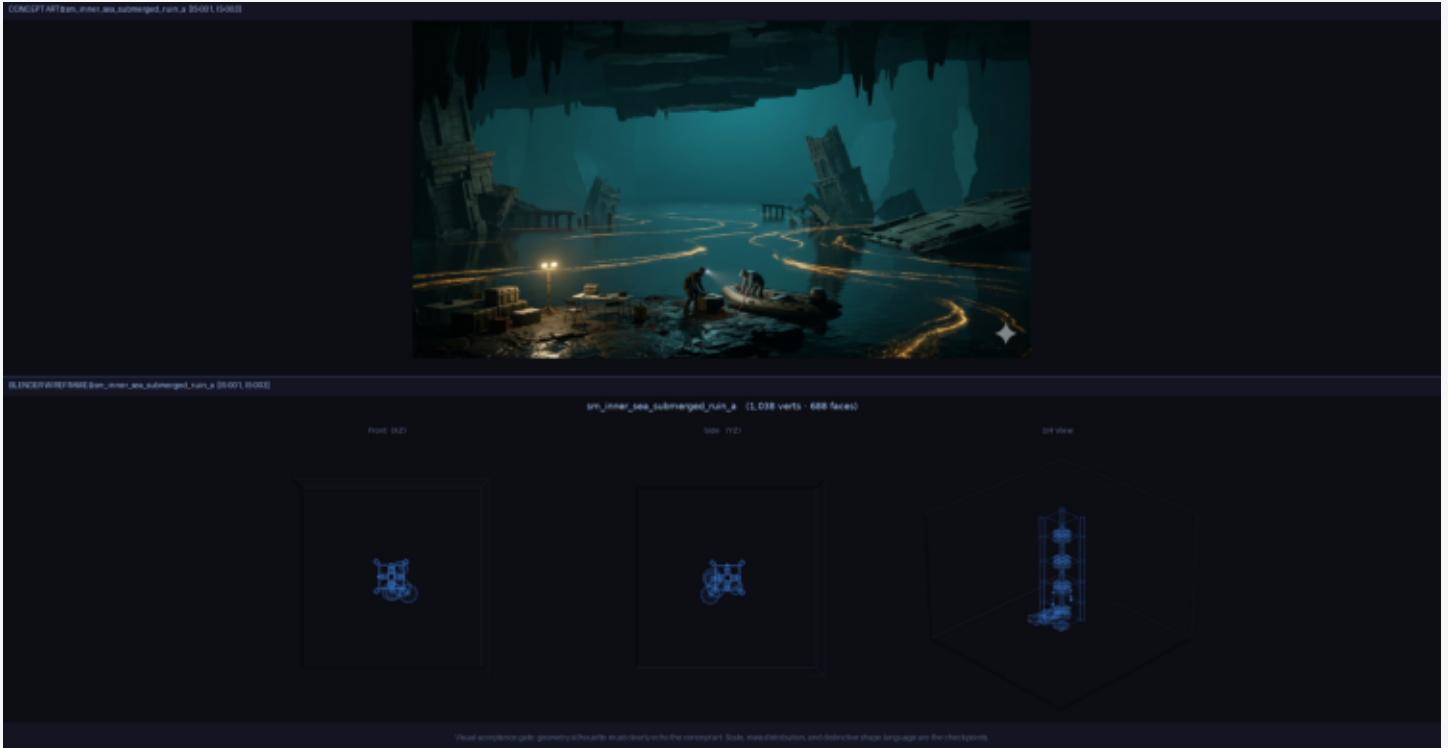
Assessment

The defining feature of GR-001 — sweeping buttress roots — was previously authored as loose vertices with no faces, rendering invisible. The rewrite sweeps five fin cross-sections along a curved path from trunk to ground as a true lofted surface, flares the 20-sided trunk with organic radius noise, and replaces the floating vein ridges with connected quad strips. Crystal accents anchor the buttress feet.

Verdict: ☐ **GO** ☐ **NO-GO (refine)**

3. Submerged Ruin — Inner Sea

Inner Sea · Concept: IS-001, IS-003



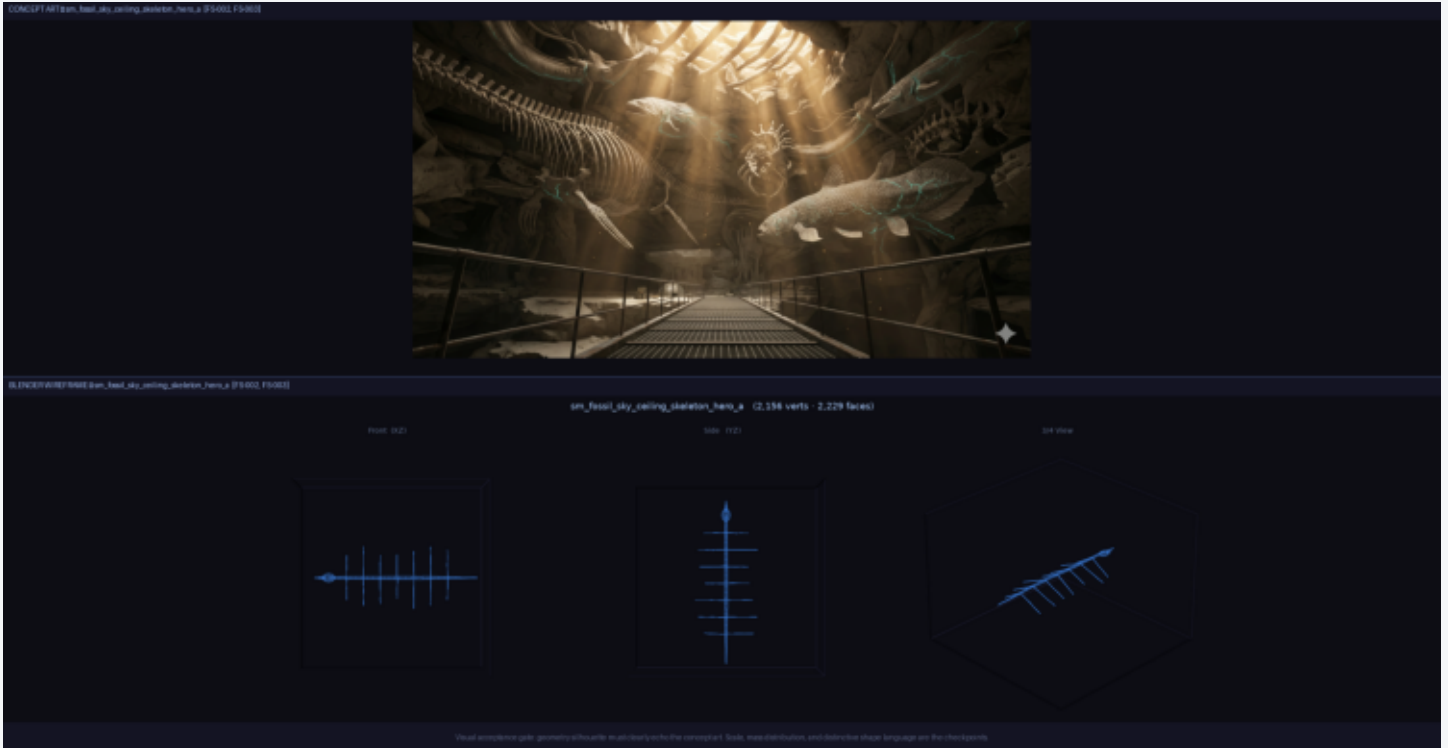
Assessment

What read as a flat machine wall panel is now ancient stone architecture: a four-sided tapered tower with gothic arched window recesses per floor, pilaster corners, a jagged broken crown, base erosion boulders and waterline barnacles. A 22 degree tilt sells the sinking, collapsing pose IS-001 frames in the dark teal sea.

Verdict: ☐ **GO** ☐ **NO-GO (refine)**

4. Ceiling Skeleton — Fossil Sky

Fossil Sky · Concept: FS-002, FS-003



Assessment

The previous skeleton was a string of unmerged icosphere blobs. The spine is now a continuous lofted tube whose radius profile swells at mid-body and tapers toward the skull, with dorsal vertebrae bumps; ribs are swept tubes following parametric arcs; the skull resolves into cranium, rostrum, jaw and eye-socket rings. The assembly is ceiling-mounted ($Z \leq 0$) so the bones hang as the concept demands.

Verdict: ☐ **GO** ☐ **NO-GO (refine)**

5. Portal Ring — Gravity Well

Gravity Well · Concept: GW-001



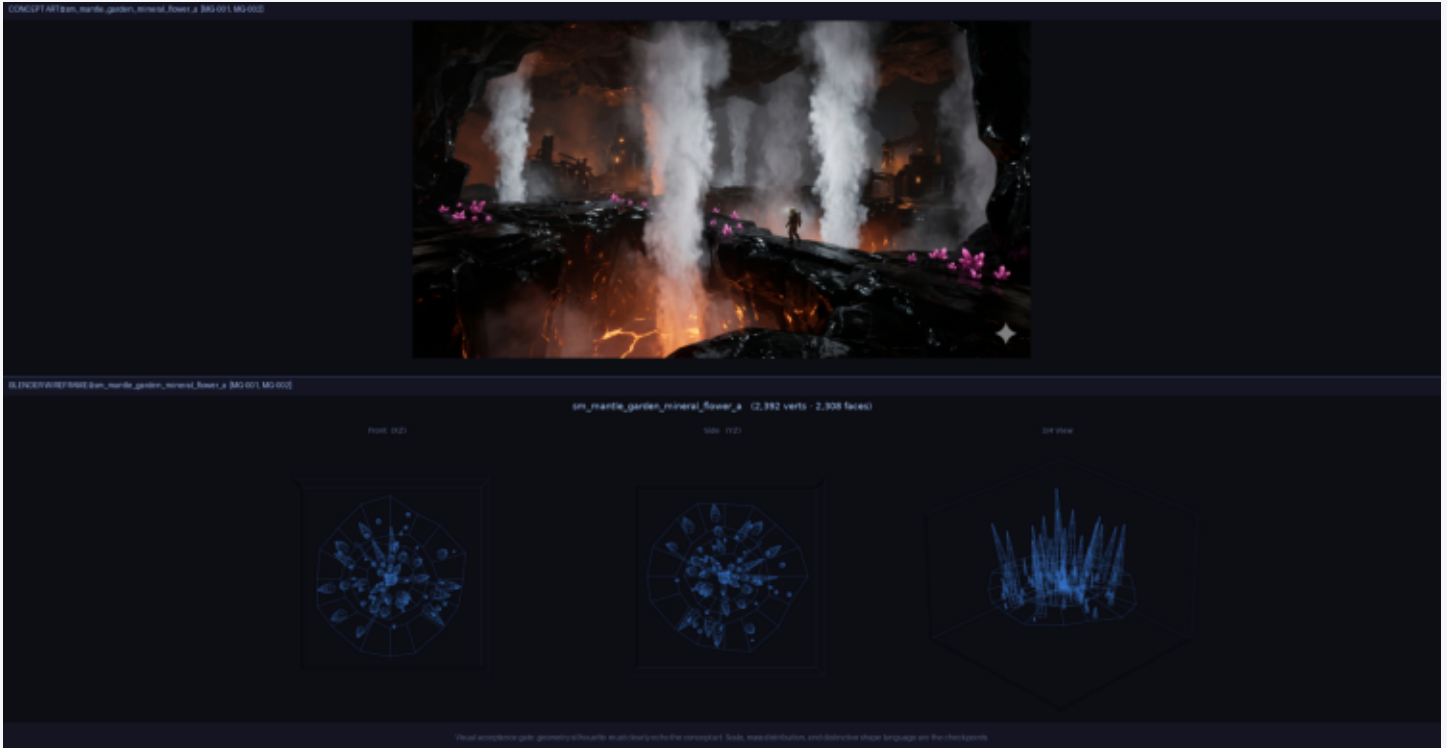
Assessment

A circular pipe torus was standing in for what GW-001 clearly shows as a rectangular stone gate. The rewrite builds two chamfered pillars and a connecting lintel, insets amber rune panels along the inner faces, seats the pillars on wide plinths, and adds triangular flanges that key the gate into the cave wall.

Verdict: ☐ **GO** ☐ **NO-GO (refine)**

6. Mineral Flower — Mantle Garden

Mantle Garden · Concept: MG-001, MG-002



Assessment

Fat hexagonal prisms became sharp faceted spires. Each spire lofts a hexagonal cross-section with a per-metre twist and an $\text{ease} = t^{1.6}$ taper down to a near-zero 0.005 m tip, clustered in a burst with outward tilt over an obsidian mound carved with heat-crack channels. Two variants ship: A (9 primary spires, 2.8 m) and B (21 primary spires, 3.5 m).

Verdict: ☐ **GO** ☐ **NO-GO (refine)**